ĐẶNG NHƯ PHƯỚC

0938361039 | phuoc.dang2104@gmail.com | Ho Chi Minh City, Vietnam

AI ML/DL ENGINEER - RESEARCHER



OVERVIEW

Github | Linkedin | E-Portfolio | LeetCode

Third-year Electrical and Electronic Engineering student at HCMUT with hands-on experience in machine learning, deep learning, and AI research. Proficient in PyTorch and TensorFlow, with expertise in computer vision, time-series modeling, explainable AI, and model optimization. Co-author of an accepted AI research paper and leader of award-winning AI projects, passionate about solving challenging AI problems through research and teamwork.

AREA OF EXPERTISE

- **Deep Learning & Neural Networks:** PyTorch, TensorFlow/Keras, CNN, MLP, LSTM, TCN, xLSTM, multi-task learning, model training, optimization, and evaluation.
- **Machine Learning & Explainable AI:** scikit-learn, LightGBM, XGBoost, Random Forest, SHAP, Integrated Gradients, feature engineering, feature selection, and hyperparameter tuning.
- **Computer Vision & AI Applications:** OpenCV, object detection, depth estimation/completion, sensor fusion, time-series classification, and signal processing.
- **AI Research & Experimentation:** Paper implementation, baseline reproduction, cross-validation, leakage-safe evaluation, ablation studies, benchmarking, and reproducible experiments.
- **Programming & AI Deployment:** Python, C/C++, SQL, Git/GitHub, Linux, Docker, ONNX, TensorFlow Lite, INT8 quantization, and FastAPI.

CERTIFICATE & ACHIEVEMENTS

- **Certifications:**
  - **IELTS 6.5**
  - Udemy Advanced Python / C / C+ + Course
  - Udemy PyTorch for Deep Learning Bootcamp
  - Udemy Full stack generative and Agentic AI Course
- **Awards and Achievements:**
  - Top **2** / 687 - FPT AIoT Challenge 2026 (**Global**)
  - Top **3** / 200+ - VNPT AI Hackathon 2025 (**Nationwide**)
  - Top **2** / 140 - FPT IoT Challenge 2025 (**Nationwide**)
  - Top **30**/500 - Green Asian Hackathon (**Global**)
  - Intel excellent innovation & impact through AI (**Global**)
  - Top **3** / 165 - HumanLog 2025 (**Nationwide**)

EXPERIENCES *(Please review my e-portfolio for more details)*

Fsoft / SILABS IoT Challenge 2025 Internship | Leader | First-runner Up Prize — 06/2026 - 09/2026

Parkinson's Intelligent Living, Long-term AIoT Remote Supervision

- Engineered a leakage-safe multimodal AI pipeline for human activity recognition, processing **2.18M sensor samples** with subject-independent evaluation across multiple datasets.
- Developed a causal signal-processing pipeline and LightGBM–TreeSHAP framework with 200 bootstrap audits, reducing **1,122 wrist–ankle features to 256**.
- Designed a compact DS-TCN (**~30K parameters, 293K MACs**), achieving **97.15% PR-AUC / 90.49% F1** for FoG detection and **87.46% Macro-F1** for fall-direction recognition.
- Trained a TENet12 CNN (**306,604 parameters**) on **12,413 audio samples** for four-class voice classification, achieving **96.37% test accuracy** and **0.9528 Macro-F1.**
- Deployed INT8 DS-TCN models on Silabs MCU, reducing model storage by **58.63%** & inference latency by **18.89%.**
- Quantized TENet12 to full-INT8 TFLite (**511.14 KiB**) with only **0.09** percentage-point accuracy loss, verifying byte-identical TFLite Micro C++ outputs on four test samples

Asian Hackathon for Green Future | Top 30/500 Global — 07/2026 - 07/2026

GreenFlow | Agentic Digital Twin Platform for Predictive Building Energy and HVAC Control

- Developed LightGBM forecasting and surrogate models on **901,824** EnergyPlus records across **308** zones, achieving **0.989 R^2** and **46.7%** lower **MAE** than persistence.
- Optimized a policy-aware MPC engine, saving **30,913 kWh (8.82%)** and cutting peak demand by **8.88%** over a 61-day replay with zero comfort or policy violations.

VNPT AI Hackathon 2025 | Leader | 2nd Runner-up Prize — 12/2025 - 12/2025

Minute - Agentic RAG, AI Meeting Workspace for BFSI Enterprise

- Developed an **Agentic RAG-powered AI meeting platform integrating real-time** VNPT GoMeet ASR, event-driven REST/WebSocket APIs, and a private VNPT Cloud backend using FastAPI, PostgreSQL, pgvector, and Docker.
- Optimized a Dockerized Vietnamese--English ASR inference service, reducing CPU usage by **75%** (**4 to 1 cores**) with approximately **3 seconds** of additional latency while maintaining the reported **99% transcription accuracy.**

Fsoft / SILABS IoT Challenge 2025 | Leader | 1st Runner-up Prize **05/2025 - 09/2025**

S.C.E.N.T (Smart Customer Experience & Edge AIoT for iNventory & Threats) - Internship Certification from FPT Software

- Built reproducible Python pipelines to collect and label **5,000+** real-world interactions across **50+** perfume SKUs, supporting ML preference modeling and model retraining.
- Developed a **BERT + spaCy NER** pipeline to extract intents and keywords from customer voice/text inputs for personalized recommendations and analytics.
- Deployed **lightweight Random Forest** models on Silicon Labs EFR32 MCUs and Raspberry Pi for recommendation and breaking/drop detection, enabling real-time inference without cloud dependency.

Intel® AI Global Impact 2025 **08/2024 - 08/2024**

Project: AIMING - AIoT Infravision for Monitoring, Inspection & Grading in Agriculture

- Designed and trained a multi-task **MobileNetV3 CNN** with four task-specific heads, achieving **91%** fruit recognition, **88%** ripeness classification, and **85%** defect detection accuracy.
- Optimized Intel hardware deployment through Keras → ONNX → OpenVINO™ IR, delivering **3×** faster inference.

HumanLog 2025 | 2nd Runner-up Prize **04/2025 - 04/2025**

Project: ESP32Cam and RFID AIoT Solution in Warehouse Distribution

- Built a lightweight **YOLO** monitoring module to detect people in restricted warehouse zones from ESP32-CAM and drone streams, sending real-time alerts to a logistics dashboard.

Stars Academy Company **06/2024 - 02/2025**

Collaborator & Teacher

- Taught Technology, Robotics, and STEM fundamentals to **classes of 40+** students across **20+ public elementary schools** in Ho Chi Minh City & Contributed to STEM and robotics curriculum development

RESEARCH & THESIS

Explainable Wrist-Based Fall Detection & Direction Recognition **Accepted**

Research Paper accepted for CSoNet 2026 | Springer LNCS Proceedings — SJR 2024 Q2

- Developed an explainable, multi-task deep learning framework for wrist-based fall detection and three-direction recognition using BITS and WEDA datasets with subject-disjoint evaluation across 10 outer test folds.
- Designed a leakage-safe, fold-specific XAI feature-selection pipeline combining Integrated Gradients, Gradient × Input, and Permutation Loss Delta to reduce **561 IMU features to 256**, improving end-to-end **Macro-F1 from 0.7669 to 0.8179** over the full-feature baseline.
- Built a compact two-head MLP with **6,653 parameters**, achieving **0.8147 Fall F1, 0.9589 Fall Recall,** and **0.8145 Direction Macro-F1,** with only **6.50 KiB** estimated INT8 weight storage.

Real-Time Dense Depth Completion for Low-Cost Autonomous Systems **In-progress**

MLIoT Lab Research Paper – aiming for A/A* Conference in Computer Vision

- Designed **JetODE**, a lightweight **583K-parameter MobileNetV4–Neural ODE architecture t**hat models local pixel-surface geometry using second-order inverse-depth jets, evolving surface value, gradients, and curvature through learned continuous dynamics and numerical integration
- Achieved **0.986** m RMSE on validation, with **50.33 ms** median inference latency using a static ODE solver, reducing latency by **32.1%** compared with adaptive integration while maintaining comparable depth accuracy.

xLSTM Direction-Excess Ranking for Vietnamese Stock Selection **3.5/4.0 Course**

Course Project | Machine Learning Fundamentals and Applications | HCMUT

- Built a Hybrid xLSTM ranking model for short-horizon stock selection, using 190,657 stock-date samples, 95 symbols, and 64-session K-line sequences to rank top-5 stocks over a 5-day horizon.
- Designed a blended scoring approach combining return/ranking, direction, and excess-return signals, and benchmarked against CNN-LSTM, TCN, PatchTST, LightGBM-style HGBR, and Kronos zero-shot models.
- Achieved best out-of-time results with IC **0.0904**, RankIC **0.0852**, Top5 Return **1.7120%**, LongShort5 **1.7462%**, with a proxy backtest return of **148.37%.**

EDUCATION

Ho Chi Minh City University of Technology - VNU **2023 - 2027**

Bachelor of Electrical and Electronic Engineering in Electrical Engineering

Relevant Courses: Probability & Statistics, Linear Algebra, Calculus I, II, Applied ML, Fundamentals of Automatic Control

MLIoT - Machine Learning & IoT Research Lab | AI Research Assistant **2025 - Present**

Participated in weekly paper reading (2 papers/week), seminars, & workshops on DL, CV, NLP, Edge AI